Paul Bittner, Ph.D.

p.m.bittner@web.de

Software engineer and researcher on programming languages; longtime professional and hobby programmer; can formalize and prove soundness properties of software; values and writes high-quality documentation; enjoys engineering expressive, easy-to-use but hard-to-misuse systems, as well as exploring problem specifications deeply and foundationally to profit from applying tools and methods across disciplines.

Software Engineer — 2025/01–present

Salzgitter Digital Solutions GmbH

Responsible for software architecture, quality assurance, testing, mentoring, AI

Research Assistant — 2020/03–2025/12

Ulm University (2020-24), Paderborn University (2024), TU Braunschweig (2025)

- 30 scientific publications (6 as first-author including 4 top-ranked conferences) in the fields of software engineering, programming languages, computer graphics
- developed several small to large-scale and peer-reviewed research prototypes
- team lead in several (international) research efforts
- supervised multiple bachelor's and master's theses within my research agenda
- teaching assistant for compiler construction, software product lines, software engineering projects, seminars

Student Research Assistant — 2019/07–2020/01

Inst. of Software Engineering and Autom. Inf., TU Braunschweig

Designed, implemented, and published [8] a meta-algorithm for reducing a particular SMT problem to SAT in the context of configuring university courses.

Student Research Assistant — 2016/11–2020/01

Computer Graphics Lab, TU Braunschweig

Responsible for maintaining and extending C++/OpenGL real-time rendering engines for research on virtual reality, panorama imagery, and animation.
I presented the lab's research to pupils once a year.

Other Positions

2014/06–2015/10 *Part-time Junior Software Developer*
Richter Datensysteme GmbH

2015/11–2017/02 *Tutor for Lectures on Algorithms and Data Structures*
Algorithms Department, TU Braunschweig

Awards

2024/10 Distinguished Artifact at OOPSLA'24 *as first author*

2024/09 Best Research Paper at SPLC'24 *as co-author*

2024/07 Best Demonstrations Paper at FSE'24 *as first author*

2022/09 Best Demonstrations Paper at SPLC'21 *as co-author*

2021/08 Best Artifact at ESEC/FSE'21 *as first author*

2016/07 Winner of the Software Engineering Student Competition at TU Braunschweig *As a team, we implemented a 3D game engine in Java including an editor and a small game. We presented our product next to all other teams at the uni's exhibition to a jury constisting of professors, company delegates, and student representatives.*

Education

PhD in Computer Science
2020–2025, Ulm University, summa cum laude
Analyzing Edits to Static Variability

Master in Computer Science
2017–2020, TU Braunschweig
Semi-Automated Inference of Feature Traceability During Software Development

Bachelor in Computer Science
2014–2017, TU Braunschweig
Evaluation of Skinning Techniques for Skeletal Animation in MonSteR

Communication

- excellent scientific writing
- high-quality documentation
- experienced speaker
- experienced supervisor and teacher

Languages & Tools

Expert: Agda, Java, C/C++, Haskell, C#, Git, LaTeX
Experienced: Python, ELisp, Linux, real-time rendering, game engine development
Knowledgable: Kotlin, JS, ANTLR, Nix/NixOS, ...

Math & Formal Methods

Expert: theorem proving, formal semantics, specify and implement PLs, DSLs, compilers, interpreters; software variability
Experienced: type theory, category theory, algebra, logic(s), constraint solving

Languages

German (native), English (fluent)

Portfolio

Vatras

Agda 12

formal framework for modeling, studying, and comparing languages for static variability in terms of proven to be correct compilers

DiffDetective

Java ×4 ×2 ×2 9

variability-aware source code differencing and large scale empirical analyses of git repositories

Feature Trace Recording

Haskell 5

concept implementation for recording code-to-feature traces from source code edits

Polypropylene

C++ 19

embedded domain-specific language and dynamic type-system for polymorphoism-aware entity component systems

PaxEngine 3

C++ 20

plugin-based 2D/3D game engine based on Polypropylene, OpenGL, SDL, Box2D, and assimp, and including several demos

Centers of Rotation Skinning

C++/GLSL 59

shaders and library to compute centers of rotation for a specific 3D skeletal animation method

A full list of my projects can be found on my website pmbittner.github.io/projects/.

Highlighted Publications

[1] Analyzing Edits to Static Variability
P. M. Bittner, DOI: 10.18725/OPARU-58798
PhD Thesis, 2025

[2] Decades of GNU Patch and Git Cherry-Pick: Can We Do Better?
A. Schultheiß, A. Boll, P. M. Bittner, S. Greiner, T. Thüm, and T. Kehrer
ICSE, 2026

[3] On the Expressive Power of Languages for Static Variability
P. M. Bittner, A. Schultheiß, B. Moosherr, J. M. Young, L. Teixeira, E. Walkingshaw, P. Ataei, and T. Thüm
PACMPL no. Object-Oriented Programming, Systems, Languages & Applications (OOPSLA), Oct. 2024, DOI: 10.1145/3689747, **Distinguished Artifact**

[4] Variability-Aware Differencing with DiffDetective
P. M. Bittner, A. Schultheiß, B. Moosherr, T. Kehrer, and T. Thüm
FSE Companion, Jul. 2024, DOI: 10.1145/3663529.3663813, **Best Demonstrations Paper**

[5] Views on Edits to Variational Software
P. M. Bittner, A. Schultheiß, S. Greiner, B. Moosherr, S. Krieter, C. Tinnes, T. Kehrer, and T. Thüm
SPLC, Aug. 2023, DOI: 10.1145/3579027.3608985,

[6] Classifying Edits to Variability in Source Code
P. M. Bittner, C. Tinnes, A. Schultheiß, S. Viegener, T. Kehrer, and T. Thüm
ESEC/FSE, Nov. 2022, DOI: 10.1145/3540250.3549108,

[7] Feature Trace Recording
P. M. Bittner, A. Schultheiß, T. Thüm, T. Kehrer, J. M. Young, and L. Linsbauer
ESEC/FSE, Aug. 2021, DOI: 10.1145/3468264.3468531, **Best Artifact**

[8] SAT Encodings of the At-Most-k Constraint – A Case Study on Configuring University Courses
P. M. Bittner, T. Thüm, and I. Schaefer
SEFM, Sep. 2019, DOI: 10.1007/978-3-030-30446-1_7

A full list of my publications can be found on my website pmbittner.github.io/publications/.